

Andrew Pols

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Education

HBSc., University of Toronto	2024 - 2028
- Specialism in Computer Science, Major in Statistics cGPA: 4.0/4.0 - Relevant coursework: Data Structures & Analysis; Software Design; Computer Organization; Data Science	

Projects

ScholarSearch – Search Engine for Research Papers	2025
- Collaborated in a 3 person team to build a research paper search engine that accelerates the research process. - Stored papers in a PostgreSQL database, along with vector embeddings in ChromaDB for scalable semantic search. - Implemented Redis caching on a database of over 1,000,000 papers to achieve a sub-200ms search response. - Containerized Postgres, ChromaDB, and Redis systems with Docker Compose for a streamlined development cycle. Tools Used: Python, FastAPI, React, Next.js, PostgreSQL, ChromaDB, Redis, Docker	
FindMySound – Music Recommender Web App	2025
- Developed a full-stack web app with Django and ReactJS that utilizes Spotify's Web API to collect user playlists. - Recommended music through the K-Nearest-Neighbours Machine Learning algorithm with Scikit-Learn. - Maintained a user base through a custom user model, with authentication through Django's Simple JWT. - Created RESTful API endpoints to handle authentication, Spotify account syncing, and recommendations. Tools Used: Python, React, Django, Django REST Framework, SQLite, Scikit-learn	
Computer Vision Sudoku – OpenCV Puzzle Processor	2024
- Developed a software that processes images and live video of a Sudoku grid to solve the puzzle. - Leveraged OpenCV classes including Hough Line Transforms to perform line-detection on a flat plane. - Performed Optical Character Recognition with Tess4JS to read each tile's digit into the solving algorithm. Tools Used: Java, Java Swing, OpenCV, Tess4JS	

Experience

Lead Robotics Programmer, FIRST Robotics	2023-2024
- Led a programming team to design, test, and deploy software for robots using Java, Swerve-Drive, & WPILib. - Used modular programming and OOP to make functional intake, elevator, and swerve subsystems. - Collaborated with mechanical and electrical subteams to ensure software matched performance expectations. - Engaged in fast-paced troubleshooting with as little as 20 minutes between matches during competition, leading the team to a provincial placement.	

Awards & Distinctions

University of Toronto Dean's List Scholar	2025
Universty of Toronto Scholar: \$10,000	2024
The Governor General's Academic Medal (Secondary School)	2024
Valedictorian, Holy Names High School	2024

Technical Skills

- Programming Languages:** Python, Java, JavaScript/JSEX/HTML/CSS, SQL, R, C
- Frameworks & Libraries:** Django REST, Node.js, ReactJS, Next.js, OpenCV, Scikit-learn, Statsmodels, Pandas
- Tools:** AWS, Git, Docker, Nginx, Gunicorn, PostgreSQL, Agile, Waterfall, Confluence, JWT